

Bitcoin & Quantum Computing

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Quantum computing

- ✦ Leverages the principles of quantum mechanics
- ✦ Allows significant speed-ups in solving specific problems
- ✦ Cryptographically Relevant Quantum Computers ,CRQCs, could break ECC-based cryptography
- ✦ To date, no quantum computer has outperformed a supercomputer on any commercially relevant application

CRQC Timeline Assessment

- ✦ US & UK - complete migration to PQC by 2035
- ✦ Several countries (including the US) target 2030 for significant progress
- ✦ Experts give >50% chance of a CRQC in the next decade
- ✦ Nvidia's CEO recently estimated it may take at least 20 years
- ✦ *"I think today that message needs to change. I think today the message needs to be: yes, unequivocally, worry about this now."* Scott Aaronson

Bitcoin Threat Model - Mining

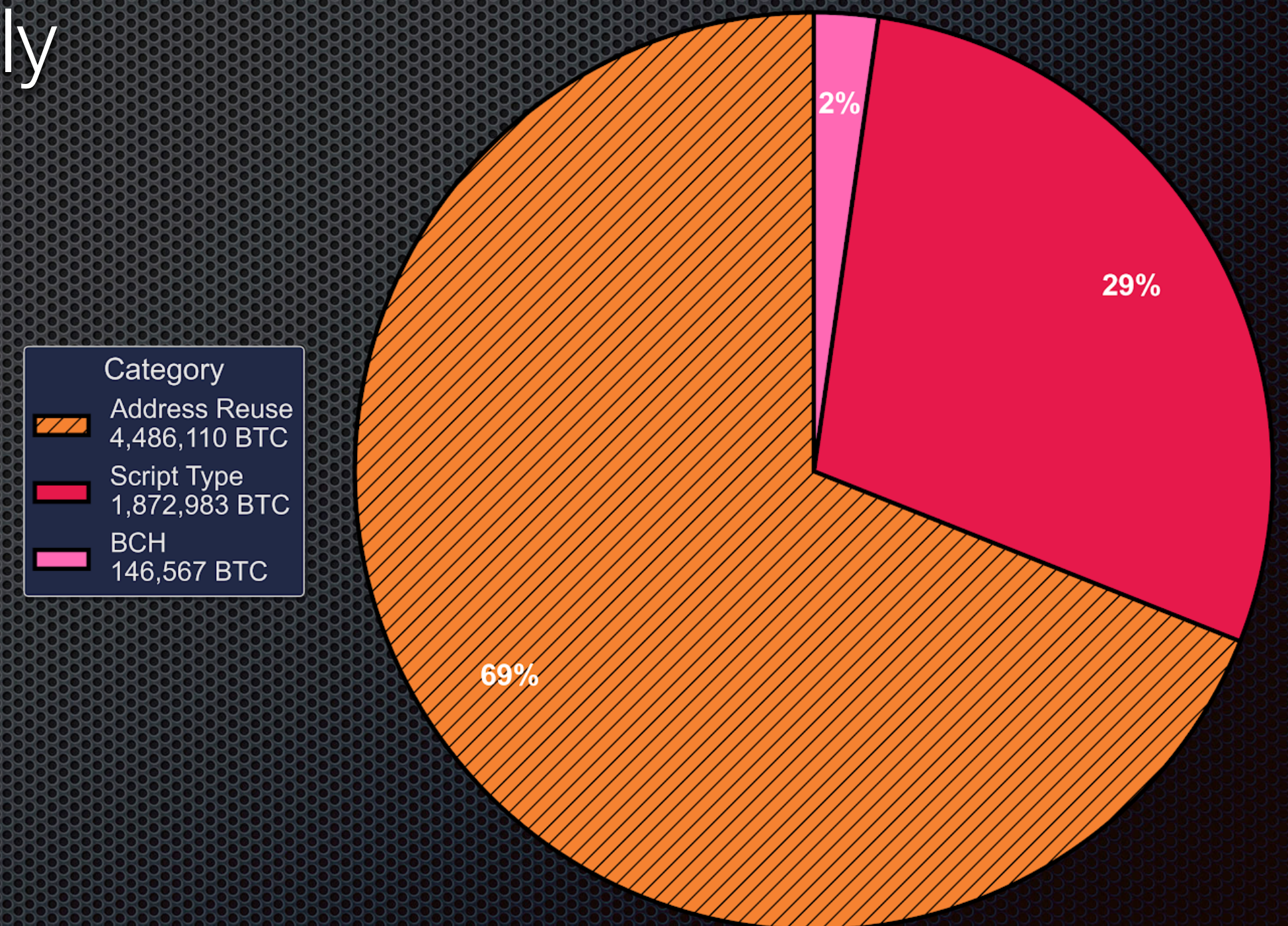
- ✦ Mining will be significantly less affected in the foreseeable future
 - ✦ Grover's Algorithm cannot be parallelized
- ✦ If quantum mining will become realistic, we can expect:
 - ✦ Significant mining centralization
 - ✦ Increase in stale blocks
 - ✦ Weakening of security assumptions (51% attack)

Bitcoin Threat Model - Signatures

- ✦ Some utxo are **immediately** vulnerable
 - ✦ P2PK, P2MS, P2TR
 - ✦ Address re-use
- ✦ All other will be vulnerable **upon spend**
- ✦ Quantum secure signatures are known, but **significantly larger**
- ✦ There is **active ongoing research** in the cryptography community

Scope of Vulnerable Funds

- ✦ Estimated ~30% of funds are immediately vulnerable
- ✦ Most are in:
 - ✦ Satoshi-era
 - ✦ Address reuse
- ✦ Exchanges could be the main targets



Immediately vulnerable ~6.51M BTC

Migration Pathways

- ✦ Migration of all utxo can take between 3 months and 2 years or more
- ✦ There are several mechanisms, some permissive and some protective:
 - ✦ CDR - Commit, Delay, Reveal
 - ✦ Hourglass strategy - Rate limit vulnerable utxo spend
 - ✦ QRAMP - Enforcing mandatory migration period

Burn vs. Steal Dilemma

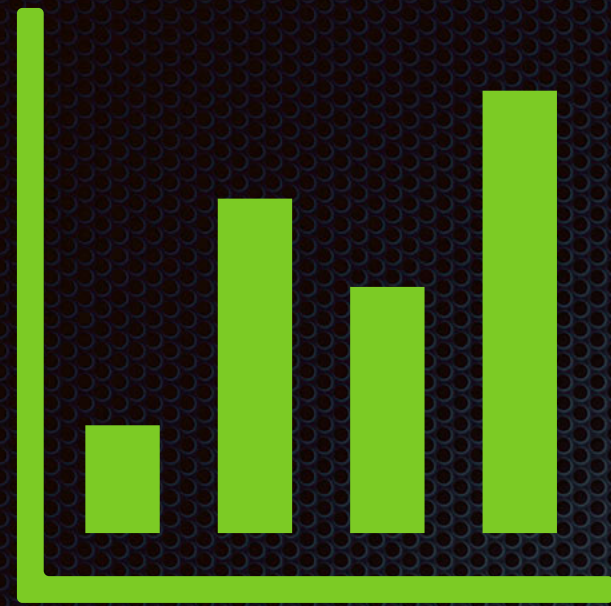
- ✦ What do we do with coins that did **not migrate**?
- ✦ **Burn** - Funds can never move
 - ✦ Philosophical clash
- ✦ **Steal** - Assets claimed by those with a CRQC
 - ✦ Destabilize the ecosystem
 - ✦ “not your keys, not your money” ↔ “your money, your keys”

Open Questions

- ✦ Financial impacts of the **Burn** vs. **Steal** dilemma
- ✦ Comparing different **migration mechanisms**
- ✦ **Signatures** (Lattice? Hash based?)

For more details

pq-bitcoin.org



PQ report

chaincode.com/bitcoin-post-quantum.pdf



Research Prize

brd.chaincode.com/#prize



Bitcoin Open Source

bosschallenge.xyz/

